

- ✓ When a colony is established, it is not so easy to eradicate; only sure method is to reach the centre of nest and kill the queen and the complementary forms.
- ✓ Before sowing granular application of Regent (Fipronil 5 % SC) @ 20 kg /ha.
- ✓ Should not be use of undecomposed green manure and FYM at termite susceptible field.
- ✓ Seed treats with Chlorpyriphos 20 EC @ 450m1/100 kg of seeds in wheat.
- ✓ In standing crop application of Chlorpyriphos 20 EC @ 4 liter/ha

❖ Cutworms

➤ Important species:

Agrostis segetum	A. epsilon
A. flammata	A. Spinifera

- **Family:** Noctuidae
- **Order:** Lepidoptera
- **Caterpillars** are damaging, Cut the seedling at ground level. Drag the seedling into the soil. Reduce the plant strand. Sometime replanting is required. Caterpillars also nibble the tubers of plants.

METHODS OF PEST CONTROL

- Any factor that is capable of making life hard for the insect that will repel or interfere with its feeding, mating, reproduction or dispersal can be taken as a method of insect control in its broadest application.

❖ Different methods of pest control / Components or tools of IPM are

- Cultural methods
- Mechanical methods
- Physical methods
- Biological methods
- Legislative methods

- Host plant resistance
- Chemical methods

❖ Cultural Methods of Pest Control

- The manipulation of cultural practices at an appropriate time for reducing or avoiding pest damage to crops is known as cultural control.
 - The cultural practices make the environment less favorable for the pests and or more favorable for its natural enemies.
 - It is the cheapest of all methods.
- (i) **Proper preparatory cultivation:** E.g., Pupae of moths, roots grubs etc.
 - (ii) **Clean cultivation:** Paddy gall fly *Orseolia oryzae* breeds on grasses such as *Panicum* sp., *Cynodon dactylon* etc. Fruit sucking moth larvae *Eudocima ancilla* on weeds of Menispermaceae.
 - (iii) **Systematic cutting and removal of infested parts:** Eg. Removal of sugarcane shoots affected by borers, Cutting and removal of infested parts of brinjal attacked by *Leucinodes orbonalis*. Clipping of tips of rice seedlings before transplanting eliminate the egg masses of stem borer. Clipping of leaf lets in coconut reduces the black headed caterpillar.
 - (iv) **Changes in the system of cultivation:** Change of banana from perennial to annual crop reduced the infestation of banana rhizome weevil *Cosmopolitus sordidus* in addition to giving increased yields. Avoiding ratoon redgram crop during offseason helps in reducing the carryover of pod fly *Melangromyza obtusa* and eriophyid mite *Aceria cajani*
 - (v) **Crop rotation:** Lady's finger followed by cotton will suffer from increased infestation of pests. Cereals followed by pulses. Cotton should be rotated with non-hosts like ragi, maize, rice to minimize the incidence of insect pests.
 - (vi) **Mixed cropping:** Intended for getting some return when one crop is attacked, the other escapes. E.g., Garden peas and Sunhemp.
 - (vii) **Growing resistant varieties:** Certain varieties resists pest attack. Eg: GEB-24 and MTU-5249 resistance to paddy BPH, Surekha variety to gall midge, TKM -6 and Ratna for stem borer.

❖ Cultural practices specially adopted for certain pests:

1. Adjusting planting or sowing or harvesting times to avoid certain pests: E.g. Early planting of paddy in *kharif* and late planting in *rabi* minimize the infestation of rice stem borer.

2. Trap cropping:

Trap Crop	Main Crop	Insect Pest
Castor	Chillies	Tobacco caterpillar, <i>Spodoptera litura</i>
Tomato	Citrus	Fruit sucking moths, <i>Otheris spp</i>
Marigold	Cotton	American bollworm, <i>Helicoverpa armigera</i>

3. Trimming field buds: Grasshopper eggs, which are laid in field bunds are destroyed by trimming field bunds.

4. Flooding the field: Flooding of fields is recommended for reducing the attack of cutworms, army worms, termites, root grubs etc., Eg: For cutworms like paddy swarming caterpillar (*Spodoptera mauritinana* and *S. exiqua*)

5. Draining the fields: In case of paddy case worm *Nymphula depunctalis* which travel from plant to plant via water. it can be eliminated by draining or drying the field.

6. Alley ways: Formation of alley ways for every 2 m in rice field reduces the BPH *Nilaparvata lugens*

❖ **Legislative / Legal / Regulatory Methods of Pest Control:** Some exotic pests introduced into our country.

Common Name	Scientific Name
Potato tuber moth	Pthorimea operculella
Cotton cushiony scale	Icerya purchasi
Woolly aphid on apple	Eriosoma lanigerum
San Jose scale	Quadraspidiotus perniciosus,
Golden cyst nematode	Globodera rostochinesis
Giant African snail	Achatina fulica (Predatory snail <i>Eugladina rosea</i>)
Serpentine leaf miner	Liriomyza trifolii
Spiralling whitefly	Alerodicus dispersus
Coconut mite	Aceria guerreoronis

❖ Quarantine:

- The word quarantine is derived from Latin word Quarantum which means ‘forty (40)’.
- The first Quarantine Act in USA came into operation in **1905**.
- While GOI passed an Act in 1914 entitled “**Destructive Insect and Pests Act of 1914**”_to prevent the introduction of any insect, fungus or other pests into our country.
- The **Directorate of Plant Protection Quarantine and Storage (DPPQS)** was established in Faridabad in 1946.
- These operates under the provisions made under the “**Destructive Insect and Pests Act of 1914**”.

- NBPGR, New Delhi for quarantine processing of all germplasm under exchange for research purposes, FRI, Dehradun for forest plants and Botanical Survey of India (BSI), Kolkata for ornamental plants to enforce quarantine laws.
- Import of potatoes from areas known to be infected with wart disease or golden cyst nematode is totally prohibited into our country.
- The consignment should accompany with the certificate issued by the Officers of agriculture department of the exporting country. This certificate is called as **‘Phytosanitary certificate’**.
- The Insecticide Act, 1968 has been enforced on 2nd September 1968 by the GOI to regulate the import, manufacture, sale, transport and distribution and use of insecticides.
- The government of India also constituted the Central Insecticide Board (CIB) to advise the state and central governments as per this act. The insecticide rules of 1971 framed under the Insecticides Act 1968 had come into force in 1971.
- ❖ **Mechanical Methods of Pest Control:** Reduction or suppression of insect pest population by means of manual devices or labour
 - (a) Hand picking and collection with hand nets and killing insects
 - (b) Provision of preventive barriers:
 - (c) **Other mechanical methods:**
 - Extraction of adult Rhinoceros beetle (*Oryctes rhinoceros*) from the crown of coconut trees using an arrow headed rod/hook.
 - Construction of rat proof godowns.
 - Use of an alkathene band around the tree trunks of mango to check the migration of first instar nymphs of mealybugs and red ants.
 - Sticky bands around tree trunks against red tree ant (*Oecophylla samaragdina*).
 - Systematic shaking of root grub adults harbored trees during evening hours to dislodge and destroy by dumping in fire.
 - Shaking of red gram plants to collect and destroy later instars of *Helicoverpa armigera*
 - Shaking the trees and bushes by which the insects fall to the ground and they can be collected.
 - Sieving and winnowing against stored grain pests.
 - Using mosquito nets fly proof cages etc.

❖ **Physical Methods of Pest Control:**

- A material called drier, consist of highly porous, finely divided silica gel which when applied abrades the insect cuticle thus encouraging loss of moisture resulting in death. It is mainly used against stored product pests.
 - Kaolinic clay after successive activation with acid and heat can be mixed with stored grain. The clay minerals absorb the lipid layer of the insect cuticle by which the insects lose their body moisture and die due to desiccation.
 - Stored products can be exposed to 55°C for 3 hours to avoid stored product pests.
 - Steam sterilization of soil kills soil insects.
 - Vapour Heat Treatment (VHT): Heated air is saturated with water (>RH 90%) for specified period of 6 to 8 hours for raising pulp temperature to 43-44.5°C in case of mango against fruit flies.
 - Oxygen stress and carbon dioxide concentration
 - **Male sterile technique:** By sterilizing the pupae of screwworm, livestock pest (*Cochliomyia hominivorax*) with radiations, sterile males were obtained. They were released @ 400/sq mile for 7 weeks.
 - Light traps are arranged for attracting the insects: Eg: Most of the moths and beetles.
 - Flame thrower is a compressed air sprayer with kerosene oil for producing flames. There is a lance, which is fitted with a burner. When the burner is heated, the kerosene oil is released, and it turns into flames. Used for burning locust populations, congregation of caterpillars, patches of weeds etc.
- ❖ **Chemical Control:** Control of insects with chemicals is known as chemical control.
 - ❖ **Pesticides:** The term pesticide is used to those chemicals which kill pests, and these pests may include insects, animals, mites, diseases or even weeds.
 - ❖ **Insecticides:** Chemicals which kill insects are called as insecticides. Insecticide may be defined as a substance or mixture of substances intended to kill, repel or otherwise prevent the insects.

DIFFERENT CLASSIFICATIONS OF INSECTICIDES

I. Based on the origin and source of supply

A. Inorganic insecticides:

- Includes arsenate and fluorine compounds.
- Sulphur as acaricides and zinc phosphide as rodenticides.

B. Organic Insecticides:

1. **Insecticides of animal origin:** Nereistoxin isolated from marine annelids, fish oil rosin soap from fishes etc.
2. **Plant Origin insecticides or Botanical insecticides:** Nicotinoids, pyrethroids, Rotenoids etc.
3. **Synthetic organic insecticides:** Organochlorines, Organophosphorus, Carbamate insecticides etc.,
4. **Hydrocarbon oils etc.**

II. Based on the mode of entry of the insecticides into the body of the insect they are groups as

1. **Contact poisons:** E.g., DDT and HCH.
2. **Stomach poisons:** Eg: Calcium arsenate, lead arsenate.

❖ Fumigants:

- A fumigant is a chemical substance which is volatile at ordinary temperatures and sufficiently toxic to the insects.
- Fumigants mostly gain entry into the body of the insect through spiracles in the trachea.
- Commonly used Fumigants and their doses:
 1. Aluminium phosphide, marketed as Celphos tablets used against field rats, groundnut bruchids etc.
 2. Carbon disulphide
 3. EDCT (Ethylene Dichloride Carbon Tetrachloride)
 4. EDB Ethylene dibromide
 5. **SO₂:** By burning sulphur in godowns SO₂ fumes are released.

❖ **Systemic insecticides:** Examples: Methyl demeton, Phosphamidon, Acephate.

❖ **Non systemic insecticides:** E.g., Chlorpyrifos, Cypermethrin, Carbaryl, Deltamethrin.